M2 Research internship proposal

Inverse problems in Stokes imaging. Application to biological imaging.

supervision: J. Flamant (CRAN) and R. Stoica (IECL)

Internship research project Imaging techniques that exploits polarization properties of light are becoming increasingly popular in biology, as *polarimetric imaging* enables many insights that are inaccessible to other conventional imaging methods [1]. Polarization information is typically described by Stokes parameters, which need to be estimated from multiple intensity measurements, defining a so-called Stokes inverse problem, see Fig. 1.

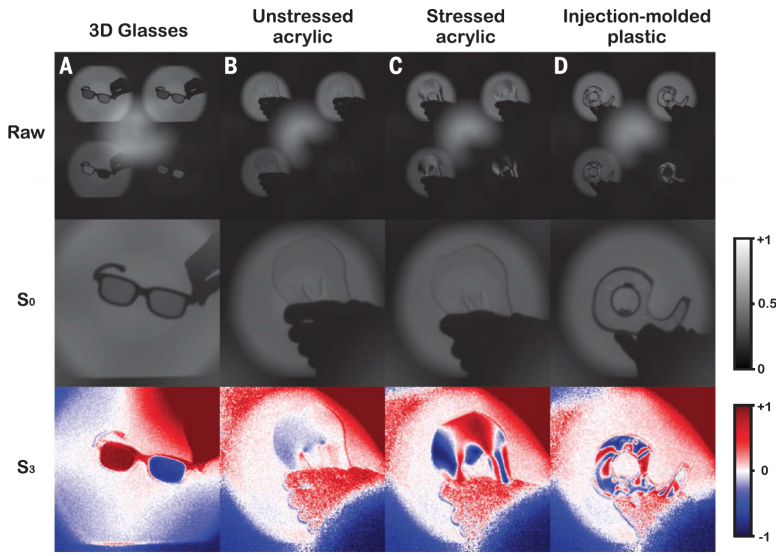

Figure 1: Reproduced from [2]. Different scenes are shown column-wise. Top: 4 intensity images to be inverted to obtain Stokes parameters; middle: standard S_0 image corresponding to conventional (polarization-insensitive) imaging; bottom: S_3 Stokes parameter (circular polarization) that reveals physical properties of the scene unaccessible to conventional imaging.

The first aim of this internship is to study the Stokes inverse problem for two noise models (additive Gaussian, Poisson) arising in imaging and develop efficient algorithms for its resolution. Two key points need to be investigated: (i) compare the merits of different measurement schemes proposed in the literature [3]–[5] w.r.t. reconstruction error and (ii) understand the role played by the Stokes physical admissibility constraint and its implementation [6]–[8] w.r.t. convergence properties. The Stokes imaging segmentation problem will then be studied and formulated withinin a Bayesian framework [9], [10], focusing on developing Markov Random Field (MRF) models [11] for polarimetric data.

The second aim of this internship will be to apply the proposed methodology to the study of the polarimetric properties of cells. Part of this work will be done in collaboration with biologists from CRAN and constitutes an exploratory research direction. This internship can be followed by a PhD position depending on the results and motivation of the candidate.

Prerequisites and funding The candidate should be enrolled in a M2 and have a M1 in either mathematics, applied mathematics or electrical engineering (signal processing), together with good oral and written english skills. The internship offers also a grant (around 550 euros per month w.r.t. the French rules). The internship is 6 months length, and it may start anytime during the period of 01/02 - 01/04/2020.

Location The candidate will be working mainly at Centre de Recherche en Automatique de Nancy (CRAN), located on the Faculté des Sciences et technologies campus in Vandoeuvre-lès-Nancy, France.

Supervision This internship falls within a collaboration between CRAN and Institut Elie Cartan de Lorraine (IECL campus de Nancy) which are located a few hundred meters apart on the campus. The candidate will be jointly supervised by researchers of both institutions:

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References

[1] V. V. Tuchin, "Polarized light interaction with tissues," *Journal of biomedical optics*, vol. 21, no. 7, p. 071 114, 2016.

[2] N. A. Rubin, G. D'Aversa, P. Chevalier, *et al.*, "Matrix fourier optics enables a compact full-stokes polarization camera," *Science*, vol. 365, no. 6448, eaax1839, 2019.

[3] M. R. Foreman and P. Török, "Information and resolution in electromagnetic optical systems," *Physical Review A*, vol. 82, no. 4, p. 043 835, 2010.

[4] M. R. Foreman, A. Favaro, and A. Aiello, "Optimal frames for polarization state reconstruction," *Physical review letters*, vol. 115, no. 26, p. 263 901, 2015.

[5] M. Born and E. Wolf, *Principles of optics: electromagnetic theory of propagation, interference and diffraction of light*, 6th. Pergamon Press, 1980.

[6] J. Birdi, A. Repetti, and Y. Wiaux, "Scalable algorithm for polarization constrained sparse interferometric stokes imaging," in *2018 IEEE 10th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, IEEE, 2018, pp. 465–469.

[7] S. Faisan, C. Heinrich, F. Rousseau, *et al.*, "Joint filtering estimation of stokes vector images based on a nonlocal means approach," *Journal of the Optical Society of America A*, vol. 29, no. 9, pp. 2028–2037, 2012.

[8] J. Flamant, S. Miron, and D. Brie, *Quaternion non-negative matrix factorization: Definition, uniqueness and algorithm*, 2019. eprint: `arXiv:1903.10593`.

[9] G. Winkler, *Image analysis, random fields and Markov chain Monte Carlo methods: a mathematical introduction*. Springer Science & Business Media, 2012, vol. 27.

[10] R. S. Stoica, A. Philippe, P. Gregori, *et al.*, "ABC shadow algorithm: A tool for statistical analysis of spatial patterns," *Statistics and computing*, vol. 27, no. 5, pp. 1225–1238, 2017.

[11] S. Z. Li, *Markov random field modeling in image analysis*. Springer Science & Business Media, 2009.